1. Write a program that is going to get n rows of m numbers. Then, the program will subtract the smallest value of every row from the row elements. Solve it using 3 functions, one for reading the array, one for subtracting the minimum and another for printing the array.

Input specification

In the beginning, you will be given 2 integers:

- the number of rows (n): an integer between 0 and 5000
- the number of numbers in every row (m): an integer between 0 and 1000 then in the following n lines you will be given m integers which are between -10000 and 10000.

Output specification

Show m integers in n lines.

Sample Input	Sample Output
3 6 5 6 1 6 1 7 10 5 15 24 13 12 9 4 16 6 7 15	4 5 0 5 0 6 5 0 10 19 8 7 5 0 12 2 3 11

2. Declare and initialize a n-by-m two-dimensional array for storing student grades. Each row of the array represents a student, and each column represents a grade on an exam the students took during the semester. Write a function that returns the lowest grade of all the students for the semester. Write a function that returns the highest grade of all the students for the semester. Write a function that returns a particular student's semester average (read from the user the index of the student). Write a function that prints the index of the highest grade. Write a function that prints the two-dimensional array in a tabular format.

Sample Input	Sample Output
3 4 77 68 86 73 96 87 89 78 70 90 86 81 Enter index: 1	Lowest grade: 68 Highest grade: 96 The average grade for student 1 is 87.50 Index of the highest grade is: 1 0

3. (Sum the major diagonal and minor diagonal in a matrix) Write a function that sums all the numbers in the major diagonal and another function that sums all the numbers in the minor diagonal in an n * n matrix of double values using the following declarations:

double sumMajorDiagonal(double m[][4])
double sumMinorDiagonal(double m[][4])

Write a test program that reads a 4-by-4 matrix and displays the sum of all its elements on the major diagonal and on the minor diagonal.

Sample Input	Sample Output
Enter a 4–by–4 matrix row by row: **1** 2 3 **4.0** 5 **6.5** **7** 8 9 **10** **11** 12 **13** 14 15 **16**	Sum of the elements in the major diagonal is 34.5. Sum of the elements in the minor diagonal is 34.0

4. (Add two matrices) Write a function to add two matrices. The declaration of the function is as follows: **void addMatrix(int r, int c, double a[r][c], double b[r][c], sum[r][c])** In order to be added, the two matrices must have the same dimensions and the same or compatible types of elements. Let c

be the resulting matrix.

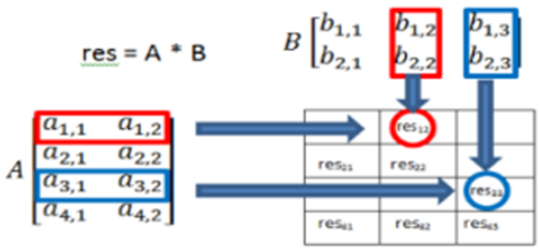
$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \\ a_{31} + b_{31} & a_{32} + b_{32} & a_{33} + b_{33} \end{pmatrix}$$

Write a test program that prompts the user to enter two n*n matrices and displays their sum.

Sample Input	Sample Output
3 3 Enter matrix1: 1 2 3 4 5 6 7 8 9 Enter matrix2: 0 2 4 1 4.5 2.2 1.1 4.3 5.2	1.0 4.0 7.0 5.0 9.5 8.2 8.1 12.3 14.2

5. When multiplying two matrices, the left matrix rows are multiplied with the right matrix columns. Every left row element is multiplied with every right column element and then summed up. For example, when the first row and the second column elements are multiplied and summed, the result is placed to the first row and the second column in the resulting matrix. Thus, inner dimensions must be the same. And the outer dimensions are the dimensions of the resulting matrix.

If:
Matrix A has 4 rows and 2 columns;
Matrix B has 2 rows and 3 columns;
Thus, matricres will have 4 rows and 3 columns;
Then, the matrix multiplication for every row and column element of the resulting array will be carried out as follows:



Question:

Write a program that is going to receive two 2D matrices and multiply them. Solve it using 2 functions, one for reading the array and another one for multiplying 2 matrices.

Input specification

First, you will be given three integers: m, n and k where the first matrix has m rows and n columns, and the second matrix has n rows and k columns and 1 < m, n , k ≤ 100. The following m rows contain n integers for the first matrix, and then, the following n rows contain k integers for the second matrix.

Output specification

Give m rows and k columns of integer numbers that represent the resulting matrix.

Sample Input I

```
2 3 2
3 4 2
1 0 1
1 4
2 0
4 4
```

Sample Input II

```
3 2 3
1 4
2 0
4 4
3 4 2
1 0 1
```

Sample Output I

```
19 20
5 8
```

Sample Output II

```
7 4 6
6 8 4
16 16 12
```

6. (Markov matrix) An n * n matrix is called a positive Markov matrix if each element is positive and the sum of the elements in each column is 1. Write the following function to check whether a matrix is a Markov matrix:

bool isMarkovMatrix (int n, double m[][n])

Write a test program that prompts the user to enter a n * n matrix of double values and tests whether it is a Markov matrix.

Sample Input	Sample Output
3 3 0.15 0.875 0.375 0.55 0.005 0.225 0.30 0.12 0.4	It is a Markov matrix.

7. The two-dimensional arrays m1 and m2 are strictly identical if their corresponding elements are equal. Write a function that returns true if m1 and m2 are strictly identical, using the following declaration:

bool equals(int n, int m, int m1[n][m], int m2[n][m])

Write a test program that prompts the user to enter two n * m arrays of integers and displays whether the two are strictly identical. Here are the sample runs:

Sample Input	Sample Output
3 3 Enter list1: 51 22 25 6 1 4 24 54 6 Enter list2: 51 22 25 6 1 4 24 54 6	The two arrays are strictly identical.

8. Suppose you need to write a program that grades multiple-choice tests. Assume there are eight students and ten questions, and the answers are stored in a two-dimensional array. Each row records a student's answers to the questions, as shown in the following array:

Students' Answers to the Questions:

	0	1	2	3	4	5	6	7	8	9
Student 0	A	B	A	C	C	D	E	E	A	D
Student 1	D	B	A	B	C	A	E	E	A	D
Student 2	E	D	D	A	C	B	E	E	A	D
Student 3	C	B	A	E	D	C	E	E	A	D
Student 4	A	B	D	C	C	D	E	E	A	D
Student 5	B	B	E	C	C	D	E	E	A	D
Student 6	B	B	A	C	C	D	E	E	A	D
Student 7	E	B	E	C	C	D	E	E	A	D

The key is stored in a one-dimensional array:

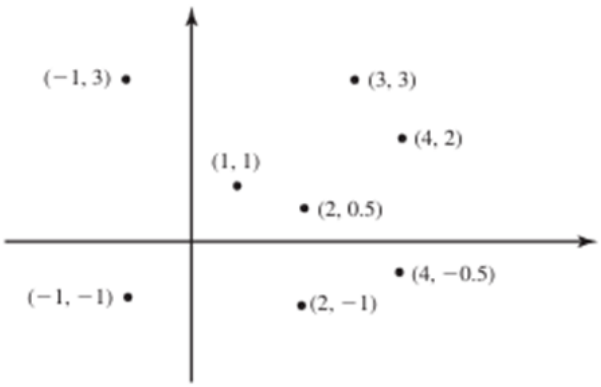
Key to the Questions:

	0	1	2	3	4	5	6	7	8	9
Key	D	B	D	C	C	D	A	E	A	D

Write a program that grades the test and displays the result. It compares each student's answers with the key, counts the number of correct answers, and displays it. Solve it using at least 2 functions.

Sample Output
Student 0's correct count is 7 Student 1's correct count is 6 Student 2's correct count is 5 Student 3's correct count is 4 Student 4's correct count is 8 Student 5's correct count is 7 Student 6's correct count is 7 Student 7's correct count is 7

9. *Given a set of points, find the two points that are nearest to each other. Compute the distances between all pairs of points and find the one with the minimum distance.



	x	y
0	−1	3
1	−1	−1
2	1	1
3	2	0.5
4	2	−1
5	3	3
6	4	2
7	4	−0.5

Sample Input	Sample Output
Enter the number of points: 8 Enter 8 points: −1 3 −1 −1 1 1 2 0.5 2 −1 3 3 4 2 4 −0.5	The closest two points are (1, 1) and (2, 0.5)